MA 324
Statistical Inference and Multivariate Analysis
IIT GUWAHATI

End-Semester Examination
14:00–17:00 IST
May 8, 2026

PART-A

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Instructions

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	5	9	19

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$\begin{aligned} Y_1 &= \theta_1 + \theta_2 + \epsilon_1, \\ Y_2 &= \theta_1 - 2\theta_2 + \epsilon_2, \\ Y_3 &= 2\theta_1 - \theta_2 + \epsilon_3, \end{aligned}$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2.

Solⁿ: Here,

$$y = \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix} \quad X = \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix} \quad \beta = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} \quad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}$$

$$\det(X^T X) = 6 \times 6 - (-3) \times (-3)$$
$$= 27 \neq 0$$

$\therefore$ $X^T X$ is invertible

$\therefore$ The LSE is given as

$$\beta = (X^T X)^{-1} X^T y$$

$$(X^T X)^{-1} = \frac{1}{\det(X^T X)} \operatorname{adj}(X^T X)$$
$$= \frac{1}{27} \begin{bmatrix} 6 & 3 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 2/9 & 1/9 \\ 1/9 & 2/9 \end{bmatrix}$$

Then, $(X^TX)^{-1}X^T$

$$= \begin{bmatrix} 2/9 & 1/9 \\ 1/9 & 2/9 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1/3 & 0 & 1/3 \\ 1/3 & -1/3 & 0 \end{bmatrix}$$

$\therefore \beta = (X^TX)^{-1}X^Ty$

$$= \begin{bmatrix} 1/3 & 0 & 1/3 \\ 1/3 & -1/3 & 0 \end{bmatrix} \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{Y_1+Y_3}{3} \\ \frac{Y_1-Y_2}{3} \end{bmatrix}$$

$\therefore$ The LSE are

$$\theta_1 = \frac{Y_1+Y_3}{3}$$

$$\theta_2 = \frac{Y_1-Y_2}{3}$$

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2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\boldsymbol{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.

   (a) (3 points) Compute the correlation matrix $\boldsymbol{\rho}$ of $\boldsymbol{X}$.

   (b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

Solⁿ: (a) Given,

$$U_1, U_2 \overset{iid}{\sim} \text{Uniform}(0,1)$$

The PDF are

$$f_{U_1}(u_1) = \begin{cases} 1 & \text{for } 0 < u_1 < 1 \\ 0 & \text{otherwise} \end{cases} \qquad f_{U_2}(u_2) = \begin{cases} 1 & \text{for } 0 < u_2 < 1 \\ 0 & \text{otherwise} \end{cases}$$

$\because$ U_1, U_2 are independent, so

$$f_{U_1,U_2}(u_1,u_2) = f_{U_1}(u_1) \cdot f_{U_2}(u_2)$$

$$= \begin{cases} 1 & \text{for } 0 < u_1, u_2 < 1 \\ 0 & \text{otherwise} \end{cases}$$

Now we have

$$E(U_1) = \frac{1}{2} = E(U_2)$$

$$\mathrm{Var}(U_1,U_2) = \mathrm{Var}(U_1) + \mathrm{Var}(U_2) \quad (\because \mathrm{Cov}(U_1,U_2) = 0)$$

$$E(U_1^2) = E(U_2^2) = \int_0^1 x^2 1 dx = \frac{x^3}{3}\Big|_0^1 = \frac{1}{3} \quad = \frac{1}{6}$$

$$E(U_1 U_2) = \int_0^1 \int_0^1 xy \cdot 1 \cdot dx dy = \frac{1}{4}$$

So, $\mathrm{Var}(U_1) = E(U_1^2) - (E(U_1))^2 = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{12} = \mathrm{Var}(U_2)$

$$\mathrm{Cov}(U_1,U_2) = E(U_1 U_2) - E(U_1)E(U_2) = \frac{1}{4} - \frac{1}{2}\cdot\frac{1}{2} = 0$$

$\rho(X_1, X_1) = \rho(X_2, X_2) = \rho(X_3, X_3) = 1$

$\rho(X_1, X_2) = \dfrac{Cov(X_1, X_2)}{\sqrt{Var(X_1)}\sqrt{Var(X_2)}} = \dfrac{Cov(U_1, U_2)}{\sqrt{Var(U_1)}\sqrt{Var(U_2)}} = 0$

$\rho(X_1, X_3) = \dfrac{Cov(U_1, U_1) + Cov(U_1, U_2)}{\sqrt{Var(U_1)}\sqrt{Var(U_1+U_2)}} = \dfrac{\frac{1}{12}}{\sqrt{\frac{1}{12}}\sqrt{\frac{1}{6}}} = \dfrac{1}{\sqrt{2}}$

$\rho(X_2, X_3) = \dfrac{1}{\sqrt{2}}$

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$$P = \begin{bmatrix} 1 & 0 & 1/\sqrt{2} \\ 0 & 1 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1 \end{bmatrix}$$

(b) * At the end of this booklet. Page - 9

3. Let the matrix of distance be

$$\begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & & & \\ 2 & 1 & 0 & & \\ 3 & 11 & 2 & 0 & \\ 4 & 5 & 3 & 3.5 & 0 \end{array}$$

(a) (6 points) Cluster the four items using each of the following procedures:

i. Complete linkage hierarchical procedure.

ii. Average linkage hierarchical procedure.

(b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

(a) i. (Complete)

Sol^n:

$$D = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & 1 & 11 & 5 \\ 1 & 0 & 2 & 3 \\ 11 & 2 & 0 & 3.5 \\ 5 & 3 & 3.5 & 0 \end{bmatrix} \end{array}$$

Min. value is of $D_{12} = 1$. So merge 1, 2

Then, $D^{(1)}_{(12)3} = \max\{D_{13}, D_{23}\} = 11$

$D^{(1)}_{(12)4} = \max\{D_{14}, D_{24}\} = 5$

$$\therefore D^{(1)} = \begin{array}{c} \\ (1,2) \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{ccc} (1,2) & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & 11 & 5 \\ 11 & 0 & 3.5 \\ 5 & 3.5 & 0 \end{bmatrix} \end{array}$$

Min. value is of $D^{(1)}_{34} = 3.5$. So merge 3, 4

Then, $D^{(2)}_{(12)(34)} = \max\{D^{(1)}_{(12)3}, D^{(1)}_{(12)4}\} = 11$

$$\therefore D^{(2)} = \begin{array}{c|cc} & (1,2) & (3,4) \\ \hline (1,2) & 0 & 11 \\ (3,4) & 11 & 0 \end{array}$$

Now all nodes 1, 2, 3, 4 get merged

$$\therefore D^{(3)} = \begin{array}{c|c} & (1,2,3,4) \\ \hline (1,2,3,4) & 0 \end{array}$$

ii. (Average)

$$D = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 1 & 11 & 5 \\ 2 & 1 & 0 & 2 & 3 \\ 3 & 11 & 2 & 0 & 3.5 \\ 4 & 5 & 3 & 3.5 & 0 \end{array}$$

Min value is of $D_{12} = 1$. So merge 1, 2.

Then, $D^{(1)}_{(12)3} = \dfrac{D_{13} + D_{23}}{2} = \dfrac{11+2}{2} = 6.5$

$D^{(1)}_{(12)4} = \dfrac{D_{14} + D_{24}}{2} = \dfrac{5+3}{2} = 4$

$$\therefore D^{(1)} = \begin{array}{c|ccc} & (1,2) & 3 & 4 \\ \hline (1,2) & 0 & 6.5 & 4 \\ 3 & 6.5 & 0 & 3.5 \\ 4 & 4 & 3.5 & 0 \end{array}$$

Min. value is of $D^{(1)}_{34} = 3.5$. So merge 3, 4

Then, $D^{(2)}_{(12)(34)} = \dfrac{D_{13} + D_{14} + D_{23} + D_{24}}{4} = \dfrac{11+5+2+3}{4} = 5.25$

Additional Pages for Answers

$$\therefore D^{(2)} = \begin{array}{c} \\ (1,2) \\ (3,4) \end{array} \begin{array}{c} \begin{array}{cc} (1,2) & (3,4) \end{array} \\ \begin{bmatrix} 0 & 5.25 \\ 5.25 & 0 \end{bmatrix} \end{array}$$

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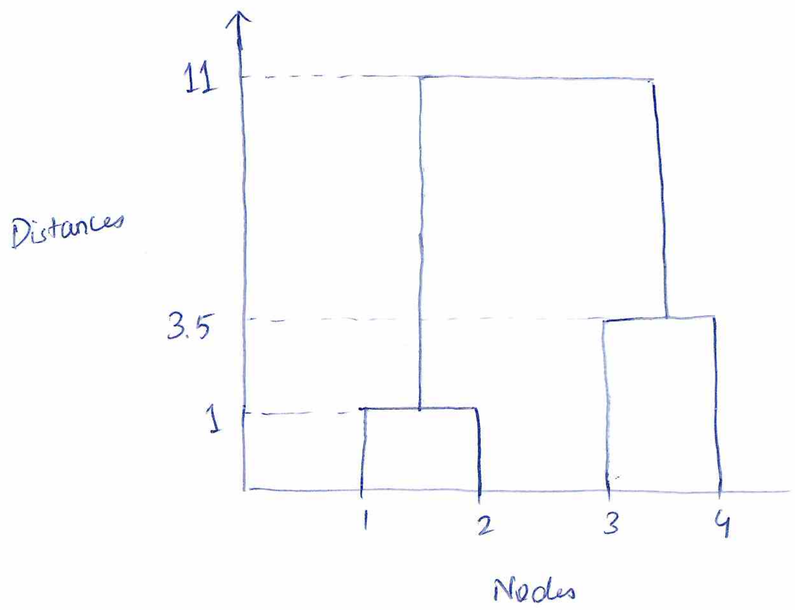
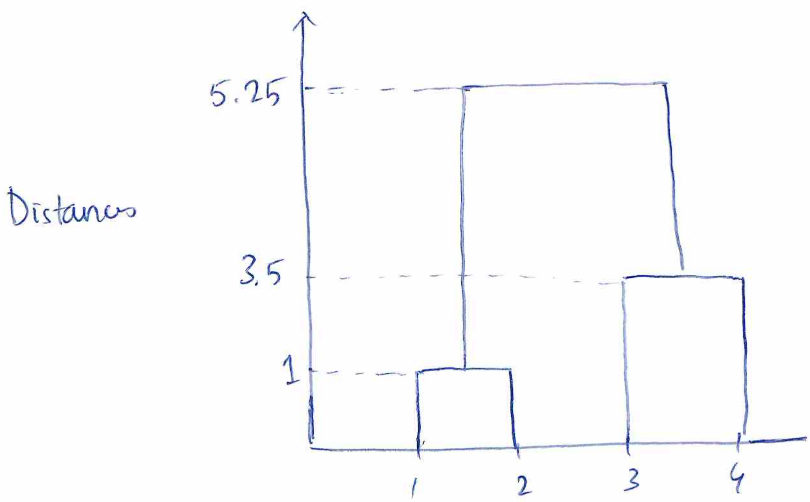
Now, all nodes 1, 2, 3, 4 get merged.

$$\therefore D^{(3)} = \begin{array}{c} \\ (1,2,3,4) \end{array} \begin{array}{c} (1,2,3,4) \\ \begin{bmatrix} 0 \end{bmatrix} \end{array}$$

(b) For complete linkage, the dendrogram is :



For average linkage, the dendrogram is



Both the linkage hierarchies have identical linkage distances for the first two iterations, as those are the original node distances. In the third and final iteration, the height in case of average linkage is lower (5.25) than that in complete linkage (11). This is because average ≤ maximum.

03

2. (b) Sol^n:

We have, $\Sigma = \begin{bmatrix} 1/12 & 0 & 1/12 \\ 0 & 1/12 & 1/12 \\ 1/12 & 1/12 & 1/6 \end{bmatrix} = \frac{1}{12}\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$

$= \frac{1}{12} \Sigma_1$

To find eigenvalues,

$$\det\left(\Sigma - \lambda_1 I\right) = 0$$

$$\Rightarrow \det\left(\frac{1}{12}\Sigma_1 - \frac{12\lambda_1}{12} I\right) = 0$$

$$\Rightarrow \det\left(\Sigma_1 - \lambda I\right) = 0 \quad \text{where } \lambda = 12\lambda_1$$

$$\Rightarrow \det\begin{pmatrix} 1-\lambda & 0 & 1 \\ 0 & 1-\lambda & 1 \\ 1 & 1 & 2-\lambda \end{pmatrix} = 0$$

$$\Rightarrow (1-\lambda)(\lambda^2 - 3\lambda + 2 - 1) + 1(0 - 1 + \lambda) = 0$$

$\Rightarrow \quad -\lambda^3+3\lambda^2-\lambda+\lambda^2-3\lambda+1-1+\lambda=0$

$\Rightarrow \quad \lambda^3-4\lambda^2+3\lambda=0$

$\Rightarrow \quad \lambda(\lambda^2-4\lambda+3)=0$

$\Rightarrow \quad \lambda(\lambda-1)\cancel{(\lambda+1)}(\lambda-3)=0$

$\therefore \quad \lambda = 3, 1, 0$

~~The largest~~

$\therefore$ Eigenvalues of Σ are $\lambda_1 = \frac{\lambda}{12} = \frac{1}{4}, \frac{1}{12}, 0$

$\therefore$ Variance of first principal component = largest eigenvalue

$= \frac{1}{4}$

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